Chapter 2

Ordinary differential equations

This chapter is not intended to be a thorough treatment of mathematical properties of ODEs. It is rather an overview of some important definitions and concepts such as problem classification, properties of linear ODE-systems with constant coefficients and stability. These matters are important for numerical analysis of methods solving ODE-problems. For a more extensive treatment of ODEs it is recommended to consult some mathematical textbook, e.g. one of those referenced to in the end of this chapter.

2.1 Problem classification

A general way of formulating a *first order scalar* ODE is

$$\frac{\mathrm{d}u}{\mathrm{d}t} = f(t,u), \tag{2.1}$$

where t is defined in some interval, bounded or unbounded, and $u(t)$ is a solution if $u(t)$ satisfies (2.1) for all t in the interval. The variable t is called the *independent* variable and u is called the *dependent* variable. The derivative $\mathrm{d}u/\mathrm{d}t$ can also be denoted $\dot{u}$ or u_t.

As we have already pointed out, the *general* solution $u = u(t,C)$ of (2.1) contains an arbitrary constant C. Hence, the general solution is a one parameter family of infinitely many solutions.

If we prescribe an initial condition $u(t_0) = u_0$ the constant C is determined and we get a unique solution, a *particular* solution.

A scalar ordinary differential equation of *second order* can be written as

$$\frac{\mathrm{d}^2u}{\mathrm{d}t^2} = f\left(t, u, \frac{\mathrm{d}u}{\mathrm{d}t}\right) \tag{2.2}$$

The general solution of (2.2) contains *two* arbitrary constants C_1 and C_2, i.e. $u = u(t,C_1,C_2)$. Therefore we need *two* conditions to determine C_1 and C_2 to get a

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unique solution. These conditions can be given in various ways. One way is to specify u at two different values of t, i.e. at $t = a$ and $t = b$:

$$u(a) = \alpha, \qquad u(b) = \beta \tag{2.3}$$

This way of specifying *boundary conditions* (BCs) to the second order ODE (2.2) is called a *boundary value* formulation, and (2.2) and (2.3) together define a *boundary value problem* (BVP). Other possible types of boundary conditions will be shown in chapter 4.

Another way to specify conditions is by *initial conditions* (ICs). Equation (2.2) can be written as a system of two first order ODEs by introducing the auxiliary variable $v = \mathrm{d}u/\mathrm{d}t$:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = v \tag{2.4a}$$

$$\frac{\mathrm{d}v}{\mathrm{d}t} = f(t, u, v) \tag{2.4b}$$

Hence, we get two differential equations of type (2.1) and a unique solution can be obtained by specifying the initial values for the two ODEs

$$u(t_0) = u_0 \qquad v(t_0) = v_0 \tag{2.5}$$

In the original formulation (2.2) the initial conditions (2.5) correspond to $u(t_0) = u_0$ and $u'(t_0) = v_0$.

Specifying ICs in this way together with the ODE-system (2.4a)–(2.4b) defines an *initial value problem* (IVP).

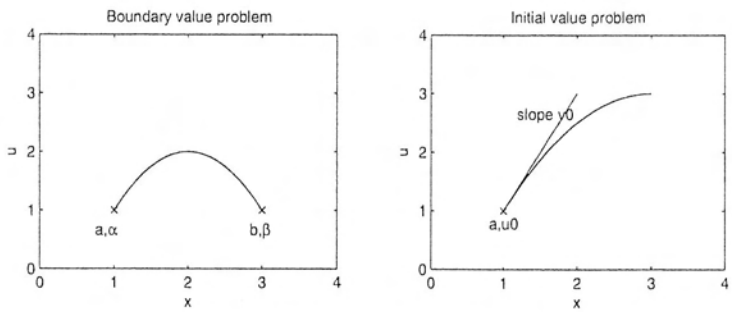


Figure 2.1: A BVP compared to an IVP

For a *system* of ODEs

$$\begin{aligned}
\frac{\mathrm{d}u_1}{\mathrm{d}t} &= f_1(t, u_1, u_2, \ldots, u_n) \\
\frac{\mathrm{d}u_2}{\mathrm{d}t} &= f_2(t, u_1, u_2, \ldots, u_n) \\
&\vdots \\
\frac{\mathrm{d}u_n}{\mathrm{d}t} &= f_n(t, u_1, u_2, \ldots, u_n)
\end{aligned} \tag{2.6}$$

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it is convenient to use vector formulation

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = \mathbf{f}(t, \mathbf{u}) \tag{2.7}$$

where the dependent variables and the right hand sides are collected in the vectors $\mathbf{u}$ and $\mathbf{f}$

$$\mathbf{u} = (u_1, u_2, \ldots, u_n)^T, \qquad \mathbf{f} = (f_1, f_2, \ldots, f_n)^T$$

Observe that there are as many dependent variables as differential equations.

The *general* solution $\mathbf{u} = \mathbf{u}(t, \mathbf{C})$, where $\mathbf{C}$ is an arbitrary constant vector in $\mathbb{R}^n$, is a family of solution curves in $[t_0, t_{\text{end}}] \times \mathbb{R}^n$, where $[t_0, t_{\text{end}}]$ is the time interval where we want the solution. If $\mathbf{C}$ is determined, e.g. by the initial condition $\mathbf{u}(t_0) = \mathbf{u}_0$ we get a unique solution, a *particular* solution. However, it is important to point out that an analytic solution of nonlinear ODE-systems is not possible to obtain in general. Numerical methods are necessary to produce graphs or tables of solutions.

The standard general formulation of an IVP is

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = \mathbf{f}(t, \mathbf{u}), \quad \mathbf{u}(t_0) = \mathbf{u}_0, \quad t \in [t_0, t_{\text{end}}] \tag{2.8}$$

and a fairly general formulation for a BVP is

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = \mathbf{f}(t, \mathbf{u}), \quad \mathbf{g}_1\big(\mathbf{u}(a)\big) = 0, \quad \mathbf{g}_2\big(\mathbf{u}(b)\big) = 0, \quad t \in [a, b] \tag{2.9}$$

where each of $\mathbf{g}_1$ and $\mathbf{g}_2$ make up n BC equations.

A scalar ordinary differential equation of *order* n

$$\frac{\mathrm{d}^n u}{\mathrm{d}t^n} = f\left(t, u, \frac{\mathrm{d}u}{\mathrm{d}t}, \frac{\mathrm{d}^2 u}{\mathrm{d}t^2}, \ldots, \frac{\mathrm{d}^{n-1} u}{\mathrm{d}t^{n-1}}\right) \tag{2.10}$$

can be written as a system of ODEs by introducing auxiliary variables $u_1 = u$, $u_2 = \mathrm{d}u/\mathrm{d}t$, $u_3 = \mathrm{d}^2u/\mathrm{d}t^2$, $\ldots$, $u_n = \mathrm{d}^{n-1}u/\mathrm{d}t^{n-1}$:

$$\begin{aligned} \frac{\mathrm{d}u_1}{\mathrm{d}t} &= u_2 \\ \frac{\mathrm{d}u_2}{\mathrm{d}t} &= u_3 \\ &\vdots \\ \frac{\mathrm{d}u_n}{\mathrm{d}t} &= f(t, u_1, u_2, \ldots, u_n) \end{aligned} \tag{2.11}$$

A particular solution is obtained either by specifying an initial vector $\mathbf{u}_0$ or by specifying n boundary conditions for u. We will not present the general case here but give different ways of specifying boundary conditions together with problems from applications in chapter 3 and 4.

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For the IVP (2.8) to have a unique solution, there is also a condition on the right hand side $\mathbf{f}(t, \mathbf{u})$ of the ODE to be differentiable with respect to t and $\mathbf{u}$. This means that f_i is differentiable with respect to t and u_j $(i, j = 1, 2, \ldots, n)$ for all $t \in [t_0, t_{\text{end}}]$ and for all points $\mathbf{u}$ in a certain domain that contains $\mathbf{u}_0$ as an interior point. This condition is a little stronger than *Lipschitz continuity*, see a mathematical textbook on ODEs.

For a BVP the question of a unique solution is more complicated. However, uniqueness for BVPs is not a big issue since many such problems occurring in applications can have several solutions which all make sense, see example 2.3 in section 2.5.

Exercise 2.1.1. Given the linear differential equation $\dot{u} = -tu$.

(a) Find the general solution of the ODE. Observe that the arbitrary constant C enters linearly in the solution.

(b) Find the particular solution satisfying the IC $u(0) = 1$.

Exercise 2.1.2. Given the nonlinear differential equation $\dot{u} = -u^2$.

(a) Give the general solution of the ODE. Observe that the arbitrary constant C enters nonlinearly in the solution.

(b) Find the particular solution satisfying $u(0) = 1$.

(c) Find the particular solution satisfying $u(0) = 0$.

Exercise 2.1.3. Given the linear second order ODE $u'' + 2u' - 3u = 0$

(a) Find the general solution of the ODE. Observe that the arbitrary constants enter linearly.

(b) Find the particular solution satisfying the IC $u(0) = 1, u'(0) = -1$.

(c) Find the particular solution satisfying the BC $u(0) = 0, u(1) = 1$.

(d) Find the particular solution satisfying the BC $u(0) = 1, u(\infty) = 0$.

Exercise 2.1.4. A particle is thrown vertically upwards. During its motion it is influenced by two forces: gravity and air resistance. The following second order ODE models the motion along the y-axis:

$$m\ddot{y} = -mg - c\dot{y}^2, \quad y(0) = y_0, \quad \dot{y}(0) = v_0$$

Rewrite this problem as a system of two ODEs.

Exercise 2.1.5. The following system of two second order ODEs models the vibrations of a mechanical system with two degrees of freedom.

$$\begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} \frac{\mathrm{d}^2\mathbf{x}}{\mathrm{d}t^2} + \begin{pmatrix} d_{\nu 1} + d_{\nu 2} & -d_{\nu 2} \\ -d_{\nu 2} & d_{\nu 2} \end{pmatrix} \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} + \begin{pmatrix} \kappa_1 + \kappa_2 & -\kappa_2 \\ -\kappa_2 & \kappa_2 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 0 \\ \hat{F}_2 \sin(\omega t) \end{pmatrix}$$

The ICs are:

$$\mathbf{x}(0) = 0, \qquad \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t}(0) = 0$$

Formulate these differential equations as a system of first order ODEs in vector form.

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Exercise 2.1.6. The following BVP models the concentration of a chemical substance in a long cylindrical catalyst pellet

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}c}{\mathrm{d}r}\right) = kc^2, \quad \frac{\mathrm{d}c}{\mathrm{d}r}(0) = 0, \quad c(R) = c_0$$

The parameters k, R and c_0 are positive parameters. The point $r = 0$ is a singular point since the left hand side term is not defined there. Use l'Hôpital's rule to find the form of the ODE at that point. Antoine l'Hôpital was French mathematician active in the second half of the 17th century.

Exercise 2.1.7. The following mixture of an ODE-system and a system of algebraic equations is called a DAE-system (Differential-Algebraic-Equations)

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathbf{f}(\mathbf{x}, \mathbf{y}), \quad \mathbf{x}(0) = \mathbf{x}_0$$

$$0 = \mathbf{g}(\mathbf{x}, \mathbf{y})$$

Assume that $\mathbf{x}$, $\mathbf{f} \in \mathbb{R}^{n_1}$ and $\mathbf{y}$, $\mathbf{g} \in \mathbb{R}^{n_2}$, where $n_1 + n_2 = n$, the number of unknowns of the system.

Formulate this DAE-system as an ODE-system with corresponding inital values assuming that the matrix $\partial\mathbf{g}/\partial\mathbf{y}$ is nonsingular.
Hint: Differentiate the algebraic system with respect to t.

This DAE-system is called an index-1 system, since a standard IVP can be obtained by differentiating part of the system only once.

2.2 Linear systems of ODEs with constant coefficients

There is one type of ODE-system that can be treated by analytical tools namely systems with constant coefficients. This problem class is very important in engineering and scientific applications, such as mechanical vibrations, electric circuits, first order chemical kinetics, etc. Linear ODE-systems with constant coefficients (LCC-systems) take the following form:

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = A\mathbf{u} + \mathbf{g}(t), \quad \mathbf{u}(0) = \mathbf{u}_0 \tag{2.12}$$

where A is a constant $n \times n$-matrix and $\mathbf{g}(t)$ is a vector of n functions of t called the *driving function.*

If $\mathbf{g}(t) \equiv 0$ the system is *homogeneous*, i.e.

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = A\mathbf{u}, \quad \mathbf{u}(0) = \mathbf{u}_0 \tag{2.13}$$

otherwise it is *inhomogeneous.*

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In the homogeneous case, $n = 1$, i.e. the scalar equation

$$\frac{\mathrm{d}u}{\mathrm{d}t} = au, \quad u(0) = u_0$$

we know that the solution is (see section 1.2)

$$u(t) = e^{at}u_0$$

where u_0 is the initial value. A generalization of this algebraic solution form to a system leads us to the following solution formula for (2.13):

$$\mathbf{u}(t) = \mathrm{e}^{At}\,\mathbf{u}_0 \tag{2.14}$$

However, what is e^{At}, e raised to the power of the matrix At?

Let's *assume* that a solution to (2.13) can be written

$$\mathbf{u}(t) = \mathbf{c}\,\mathrm{e}^{\lambda t} \tag{2.15}$$

where $\mathbf{c}$ and λ are to be determined. Inserting this expression into (2.13) gives the following relation to be fullfilled for all t:

$$\mathbf{c}\lambda\,\mathrm{e}^{\lambda t} = A\mathbf{c}\,\mathrm{e}^{\lambda t}$$

This leads to the *eigenvalue problem*:

$$A\mathbf{c} = \lambda\mathbf{c} \qquad \text{or} \qquad (A - \lambda I)\mathbf{c} = 0 \tag{2.16}$$

We are interested in solutions where $\mathbf{c} \neq 0$ ($\mathbf{c} = 0$ gives the trivial solution $\mathbf{u} = 0$). This is possible only if

$$\det(A - \lambda I) = 0 \tag{2.17}$$

This equation is called the *characteristic equation* and its roots are known as the *eigenvalues* of the matrix A. If A is an $n \times n$-matrix, (2.17) is a polynomial equation of degree n, and consequently has n roots $\lambda_1, \lambda_2, \ldots, \lambda_n$. The roots can be real or complex.

When an eigenvalue λ_i is known we compute the corresponding *eigenvector* $\mathbf{c}_i$ by solving the homogeneous linear equation system

$$(A - \lambda_i I)\mathbf{c}_i = 0 \tag{2.18}$$

The solution $\mathbf{c}_i$ is not unique; $\mathbf{c}_i$ is determined only up to a scalar factor. Also $\mathbf{c}_i$ can be complex even if A is real. If A is real and symmetric, however, both λ_i and $\mathbf{c}_i$ are real.

From now on we assume that the n eigenvectors of the matrix A are linearly independent. Such a matrix is called *diagonalizable* or *non-defective*. (If A has fewer than n linearly independent eigenvectors it is called *defective*). For a non-defective matrix the relation (2.18) can be written in the form

$$A\mathbf{c}_i = \lambda_i\mathbf{c}_i \rightarrow AS = S\Lambda \rightarrow A = S\Lambda S^{-1} \quad \text{or} \quad S^{-1}AS = \Lambda$$

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where S is a matrix composed of the eigenvectors as columns and Λ is a diagonal matrix of the eigenvalues:

$$S = (\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_n), \qquad \Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$$

Going back to (2.15) we now have n linearly independent solutions to (2.13):

$$\mathbf{u}_i(t) = \mathbf{c}_i \, \mathrm{e}^{\lambda_i t}, \quad i = 1, 2, \ldots, n \tag{2.19}$$

Since (2.13) is a linear homogeneous problem, the solutions (2.19) can be superposed, i.e. linearly combined to a *general solution*:

$$\mathbf{u}(t) = \sum_{i=1}^{n} \alpha_i \mathbf{c}_i \, \mathrm{e}^{\lambda_i t} \tag{2.20}$$

where α_i are arbitrary constants. Formula (2.20) can also be written in matrix form:

$$\mathbf{u}(t) = S \, \mathrm{e}^{\Lambda t} \, \alpha \tag{2.21}$$

$\mathrm{e}^{\Lambda t}$ is a diagonal matrix with $\mathrm{e}^{\lambda_i t}$ as diagonal elements:

$$\mathrm{e}^{\Lambda t} = \mathrm{diag}(\mathrm{e}^{\lambda_1 t}, \mathrm{e}^{\lambda_2 t}, \ldots, \mathrm{e}^{\lambda_n t})$$

and α is an arbitrary column vector $(\alpha_1, \alpha_2, \ldots, \alpha_n)^T$. When the initial vector $\mathbf{u}(0) = \mathbf{u}_0$ is known we get by inserting $t = 0$ into (2.21):

$$\mathbf{u}(0) = S\alpha = \mathbf{u}_0 \quad \rightarrow \quad \alpha = S^{-1}\mathbf{u}_0$$

and (2.21) can be written

$$\mathbf{u}(t) = S \, \mathrm{e}^{\Lambda t} \, S^{-1} \mathbf{u}_0 \tag{2.22}$$

Hence comparing with (2.14) we define e^{At}, the *exponential matrix*, as

$$\mathrm{e}^{At} = S \, \mathrm{e}^{\Lambda t} \, S^{-1} \tag{2.23}$$

The expression (2.23) can be used only if S has an inverse (S is non-singular). It can be shown that if all eigenvalues of A are different then A is non-defective. This is true also when A is symmetric.

However there exist matrices which are defective. In that case the formula (2.22) does not make sense since S has no inverse, but we can always use the Taylor series expansion as definition of e^{At}:

$$\mathrm{e}^{At} = I + At + \frac{t^2}{2!}A^2 + \frac{t^3}{3!}A^3 + \cdots + \frac{t^n}{n!}A^n + \ldots \tag{2.24}$$

It can be shown that if S has an inverse, the two definitions (2.23) and (2.24) are equivalent.

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If the ODE-system is LCC and inhomogeneous, i.e. the right hand side contains a driving function $\mathbf{g}(t)$ as in equation (2.12)

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = A\mathbf{u} + \mathbf{g}(t), \quad \mathbf{u}(0) = \mathbf{u}_0$$

we obtain the analytic solution in the form of an integral, the Duhamel formula:

$$\mathbf{u}(t) = \mathrm{e}^{At}\,\mathbf{u}_0 + \int_0^t \mathrm{e}^{A(t-\tau)}\,\mathbf{g}(\tau)\,\mathrm{d}\tau \tag{2.25}$$

In case the driving vector function $\mathbf{g}(t)$ is a constant vector $\mathbf{g}$ the integral can be evaluated and the solution of the ODE-system is (if A has an inverse)

$$\mathbf{u}(t) = \mathrm{e}^{At}\,\mathbf{u}_0 + A^{-1}(\mathrm{e}^{At} - I)\mathbf{g} \tag{2.26}$$

Exercise 2.2.1. Compute the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} -2 & 1 \\ -4 & 3 \end{pmatrix}$$

Exercise 2.2.2. Compute e^{At} for the matrix given in 2.2.1.

Exercise 2.2.3. Solve the differential equation $\dot{\mathbf{u}} = A\mathbf{u}, \quad \mathbf{u}_0 = (1,1)^T$, where A is the matrix in 2.2.1, using the result obtained in 2.2.2.

Exercise 2.2.4. Compute eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 0 & 1 \\ -5 & 2 \end{pmatrix}$$

Exercise 2.2.5. Solve the differential equation $\dot{\mathbf{u}} = A\mathbf{u}, \quad \mathbf{u}_0 = (1,1)^T$, where A is the matrix in 2.2.4.

Exercise 2.2.6. Verify that the matrix

$$B = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

is defective

Exercise 2.2.7. Compute e^{Bt} for the matrix given in 2.2.6.

Exercise 2.2.8. Verify the solution formula (2.25) for the inhomogeneous LCC-problem (2.12).

Exercise 2.2.9. Modify the formulas (2.14) and (2.25) to the case where the initial value is $\mathbf{u}(t_0) = \mathbf{u}_0$, i.e. $\mathbf{u}$ is known in the initial point $t = t_0$.

Exercise 2.2.10. In what sense is the system (2.13) and its solutions linear?

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2.3 Some stability concepts for ODEs

The goal of stability analysis is to investigate if small perturbations in a system will cause large changes in the evolution of the system, in which case the system is *unstable*. In this chapter we present two definitions of stability for ODE-systems. The first definition refers to what is meant by a stable *solution trajectory* and the second concerns the stability of the special solutions called *critical points* (also called equilibrium points). In this textbook let's call these two stability concepts *analytical stablity*.

Another stability concept deals with the stability of numerical solutions. It is important to distinguish between analytical stability and *numerical stability*. Numerical stability is introduced in chapter 3.

2.3.1 Stability for a solution trajectory of an ODE-system

As already mentioned analytic solution of a nonlinear ODE-system (2.7) is not possible in general. However for small perturbations the stability analysis can be based on a linear approximation leading to an LCC-system. First we need a definition of stability. In the definition below we restrict ourselves to *autonomous* systems, i.e. ODE-systems of the form

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = \mathbf{f}(\mathbf{u}) \tag{2.27}$$

Definition 2.1. A solution $\mathbf{u}(t)$ of (2.27) is *stable* if every solution which starts sufficiently close to $\mathbf{u}(t)$ at $t = t_0$ remains close to $\mathbf{u}(t)$ for all $t > t_0$.
The solution $\mathbf{u}(t)$ is *unstable* if there exists at least one solution starting close to $\mathbf{u}(t)$ at $t = t_0$ which does not remain close to $\mathbf{u}(t)$ for all $t > t_0$. ◇

The stability question can be completely answered for LCC-systems

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = A\mathbf{u} \tag{2.28}$$

Consider first the scalar ODE

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \lambda u \tag{2.29}$$

Assume that $u(t)$ is a solution of (2.29). At the point (t_0, u_0) on the solution trajectory a small perturbation δu_0 takes us to a new solution trajectory $u(t) + \delta u(t)$ starting at $(t_0, u_0 + \delta u_0)$. This trajectory is followed for $t > t_0$. Since the perturbed solution satisfies the ODE

$$\frac{\mathrm{d}(u + \delta u)}{\mathrm{d}t} = \lambda(u + \delta u) \tag{2.30}$$

which can be reduced to an ODE for the perturbation only:

$$\frac{\mathrm{d}\delta u}{\mathrm{d}t} = \lambda \delta u, \quad \delta u(t_0) = \delta u_0 \tag{2.31}$$

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Since the constant λ may be complex, i.e. $\lambda = \mu + \mathrm{i}\omega$, where $\mathrm{i} = \sqrt{-1}$, the solution of the scalar ODE (2.31) can be written as

$$\delta u(t) = \mathrm{e}^{\lambda t}\,\delta u_0 = \mathrm{e}^{\mu t}\,\mathrm{e}^{\mathrm{i}\omega t}\,\delta u_0 = \mathrm{e}^{\mu t}(\cos(\omega t) + \mathrm{i}\sin(\omega t))\delta u_0 \tag{2.32}$$

The only part in the expression (2.32) that can increase without bounds as t increases is $\mathrm{e}^{\mu t}$. Hence, we see that the solution is stable iff $\mu = \mathrm{Re}(\lambda) \leq 0$. To be more specific, if $\mathrm{Re}(\lambda) < 0$, the perturbation $\delta u(t) \to 0$ as $t \to \infty$. This is called *asymptotic stability*. If $\mathrm{Re}(\lambda) = 0$, the perturbation $\delta u(t)$ will stay close to $u(t)$ as $t \to \infty$ but $\delta u(t)$ will not vanish, see the figure below where a perturbation is introduced at the point marked o.

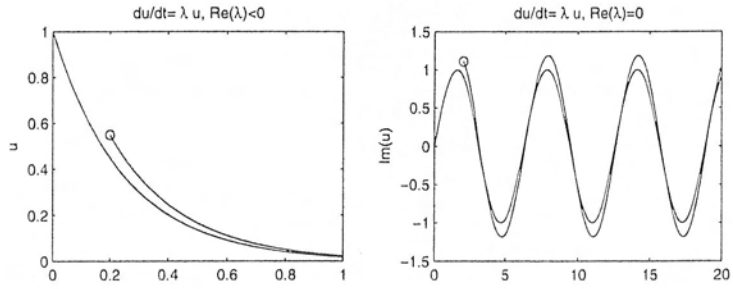


Figure 2.2: Asymptotic stability and stability

A similar analysis of (2.28) gives the following LCC-system for the perturbation $\delta\mathbf{u}$

$$\frac{\mathrm{d}\delta\mathbf{u}}{\mathrm{d}t} = A\delta\mathbf{u}, \quad \delta\mathbf{u}(t_0) = \delta\mathbf{u}_0 \tag{2.33}$$

From the explicit solution formula (2.20) it is clear that stability depends on the eigenvalues $\lambda_i, i = 1, 2, \ldots, n$ of the matrix A. Hence the LCC-system (2.33) is asymptotically stable if $\mathrm{Re}(\lambda_i) < 0$ for $i = 1, 2, \ldots n$.

If $\mathrm{Re}(\lambda_i) = 0$ the system is stable if every λ_i is *simple*. If $\mathrm{Re}(\lambda_i) = 0$ and one or more of the λ_i are *multiple* eigenvalues the stability problem is more intricate. The solution can be stable or unstable and each case must be investigated separately. We will not treat this case here (see exercise 2.3.7 for an example) but refer to a mathematical textbook on ODEs.

If we plot the eigenvalues of A in the complex plane, asymptotic stability means that the eigenvalues are situated in the *left half* of the complex plane.

Example 2.1. The eigenvalues of the matrices

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 3 & 1 & -2 \\ 2 & 2 & -2 \end{pmatrix} \qquad B = \begin{pmatrix} -1 & 0 & 0 \\ 3 & 1 & -2 \\ 2 & 2 & 1 \end{pmatrix}$$

have the following location in the complex plane

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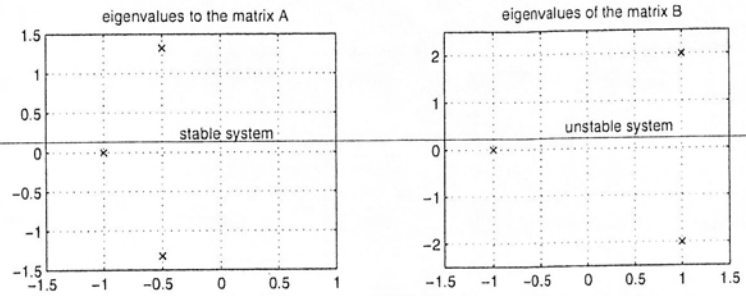


Figure 2.3: Eigenvalues in the complex plane

For the nonlinear autonomous system (2.27), the stability of a solution can be examined if the system is approximated by an LCC-system. Assume that $\mathbf{u}(t)$ is a solution of (2.27). At time $t = t_0$, where $\mathbf{u} = \mathbf{u}(t_0)$, a small perturbation $\delta\mathbf{u}_0$ brings us to a new solution trajectory, $\mathbf{u}(t) + \delta\mathbf{u}(t)$, which is followed for $t > t_0$. The perturbed solution satisfies the ODE-system

$$\frac{\mathrm{d}(\mathbf{u} + \delta\mathbf{u})}{\mathrm{d}t} = \mathbf{f}(\mathbf{u} + \delta\mathbf{u}) \tag{2.34}$$

Taylor expansion of the right hand side up to the first order term gives

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} + \frac{\mathrm{d}\delta\mathbf{u}}{\mathrm{d}t} = \mathbf{f}(\mathbf{u}) + \frac{\partial \mathbf{f}}{\partial \mathbf{u}}\delta\mathbf{u} + \mathcal{O}\left(\|\delta\mathbf{u}\|^2\right) \tag{2.35}$$

In (2.35) the *jacobian* matrix is introduced

$$J(\mathbf{u}) = \frac{\partial \mathbf{f}}{\partial \mathbf{u}} = \begin{pmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} & \cdots & \frac{\partial f_1}{\partial u_n} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} & \cdots & \frac{\partial f_2}{\partial u_n} \\ \vdots & \vdots & & \vdots \\ \frac{\partial f_n}{\partial u_1} & \frac{\partial f_n}{\partial u_2} & \cdots & \frac{\partial f_n}{\partial u_n} \end{pmatrix} \tag{2.36}$$

Since $\mathbf{u}$ is a solution of (2.27) we obtain, after neglecting the higher order terms, the (linearized) *variational equation*

$$\frac{\mathrm{d}\delta\mathbf{u}}{\mathrm{d}t} = J(\mathbf{u})\delta\mathbf{u}, \quad \delta\mathbf{u}(t_0) = \delta\mathbf{u}_0 \tag{2.37}$$

If the jacobian is frozen at $t = t_0$ to a constant matrix J_0, i.e.

$$J_0 = \frac{\partial \mathbf{f}}{\partial \mathbf{u}}(\mathbf{u}(t_0)) \tag{2.38}$$

we can use the stability results for LCC-systems to answer questions about the stability of solution trajectories in a close neighbourhood of the given solution trajectory.

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2.3.2 Stability for critical points of ODE-systems

For an autonomous nonlinear ODE-system (2.27) it is of special interest to investigate the stability at *critical points*, i.e. points satisfying the algebraic equation

$$\mathbf{f}(\mathbf{u}) = 0 \tag{2.39}$$

Critical points are also called the *stationary* or *steady-state* solutions, since such a solution does not change with time t.

Let $\mathbf{u}^*$ be a critical point of (2.39). We want to investigate if this point is stable. The stability concept has to be modified a little for a critical point, since it is a constant vector.

Definition 2.2. Assume the critical point $\mathbf{u}^*$ is perturbed to a solution $\mathbf{u}^*+\delta\mathbf{u}(t)$ in the neighbourhood of $\mathbf{u}^*$. If all perturbed solutions converge to the critical point $\mathbf{u}^*$ this point is said to be asymptotically stable. If at least one perturbed solution diverges away from the critical point it is unstable. ◇

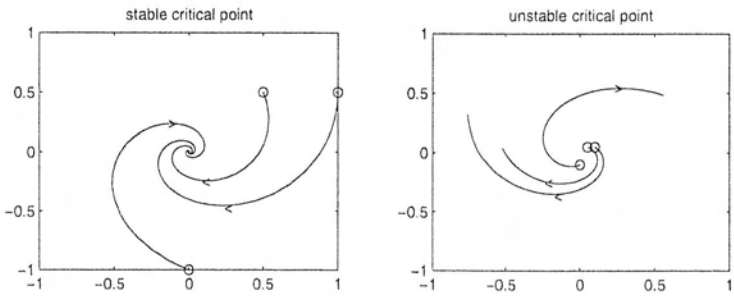


Figure 2.4: Stability of critical points

If the perturbation $\delta\mathbf{u}$ is small we can approximate $\mathbf{f}(\mathbf{u})$ close to the point $\mathbf{u}^*$ by using Taylor's expansion up to the first order term:

$$\mathbf{f}(\mathbf{u}^* + \delta\mathbf{u}) = \mathbf{f}(\mathbf{u}^*) + J(\mathbf{u}^*)\delta\mathbf{u} + \mathcal{O}\Big(\|\delta\mathbf{u}\|^2\Big) \tag{2.40}$$

The perturbed solution satisfies the corresponding differential equation, i.e.

$$\frac{\mathrm{d}\mathbf{u}^*}{\mathrm{d}t} + \frac{\mathrm{d}\delta\mathbf{u}}{\mathrm{d}t} = \mathbf{f}(\mathbf{u}^*) + J(\mathbf{u}^*)\delta\mathbf{u} \tag{2.41}$$

where we have neglected the higher order terms in (2.40). Since $\mathbf{u}^*$ is a constant and $\mathbf{f}(\mathbf{u}^*) = 0$ we get an ODE-system, the variational equation, for the perturbation term $\delta\mathbf{u}$:

$$\frac{\mathrm{d}\delta\mathbf{u}}{\mathrm{d}t} = J(\mathbf{u}^*)\delta\mathbf{u} \tag{2.42}$$

This is an LCC-system of type (2.28). The perturbed solution will converge to the critical point $\mathbf{u}^*$ if the eigenvalues of $J(\mathbf{u}^*)$ are situated in the left half of

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the complex plane, so the behaviour of solutions close to a critical point can be described in the same way as the solutions of an LCC-problem.

Hence to investigate the stability at a critical point, proceed as follows:

1) Compute a critical point $\mathbf{u}^*$ by solving $\mathbf{f}(\mathbf{u}) = 0$ with algebraic tools or numerically with Newton's method (see appendix A.1)

2) Form the jacobian $J(\mathbf{u})$ and compute $J^* = J(\mathbf{u}^*)$

3) Compute the eigenvalues λ_i of J^* and check if $\mathrm{Re}(\lambda_i) < 0$ for all i.

Example 2.2. A famous ODE-system called the *Lotka-Volterra* differential equations or the *predator-prey* model is

$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t} &= ax - bxy \\ \frac{\mathrm{d}y}{\mathrm{d}t} &= -cy + dxy \end{aligned}$$

where a, b, c and d are positive parameters.

The critical points of this system are $(0,0)$ and $(c/d, a/b)$. The jacobian of the system is

$$J = \begin{pmatrix} a - by & -bx \\ dy & -c + dx \end{pmatrix}$$

and the values of the jacobian at the critical points are

$$J(0,0) = \begin{pmatrix} a & 0 \\ 0 & -c \end{pmatrix}, \qquad J(c/d, a/b) = \begin{pmatrix} 0 & -bc/d \\ da/b & 0 \end{pmatrix}$$

The eigenvalues of $J(0,0)$ are a and $-c$, i.e. one eigenvalues is positive and the critical point $(0,0)$ is thus unstable.

The eigenvalues of $J(c/d, a/b)$ are $\pm \mathrm{i}\sqrt{ac}$. They have zero real parts, but are simple which implies that the linearized system at the critical point $(c/d, a/b)$ is stable.

Exercise 2.3.1. Given the following ODE-system

$$\dot{\mathbf{u}} = A\mathbf{u}, \qquad \mathbf{A} = \begin{pmatrix} -2 & 1 \\ -4 & 3 \end{pmatrix}$$

Find the critical points and characterize the stability properties at the critical points.

Exercise 2.3.2. What is the critical point $\mathbf{x}_{\mathrm{cr}}$ of the ODE-system (A is nonsingular)?

$$\dot{\mathbf{x}} = A\mathbf{x} - \mathbf{b}$$

Introduce the variable $\mathbf{y} = \mathbf{x} - \mathbf{x}_{\mathrm{cr}}$. Derive the ODE for $\mathbf{y}$. This shows that stability does not depend on $\mathbf{b}$.

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Exercise 2.3.3. Find the critical points and the solution of the ODE-system

$$\frac{\mathrm{dw}}{\mathrm{d}t} = k_t \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \mathrm{w} + k_i \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathrm{w}(0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The parameters k_t and k_i are both positive. What about the stability at the critical points?

Exercise 2.3.4. Given the following ODE

$$\frac{\mathrm{d}u}{\mathrm{d}t} = 1 - u^2$$

(a) Find the critical points of the ODE

(b) Decide for each critical point if the solution is stable or not.

Exercise 2.3.5. Given the second order ODE, known as van der Pol's equation

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \epsilon(x^2 - 1)\frac{\mathrm{d}x}{\mathrm{d}t} + x = 0$$

where the parameter $\epsilon > 0$.

(a) Find the critical point of the ODE-system

(b) Is the solution at the critical point stable or unstable? Hint: Rewrite as a system of first order ODEs.

Exercise 2.3.6. Consider the homogenous vibration equation (see section 1.3)

$$m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + d_\nu\frac{\mathrm{d}y}{\mathrm{d}t} + ky = 0$$

The parameters m, d_ν and k are positive.

(a) Find the critical point.

(b) Sketch how the eigenvalues at the critical point 'move' in the complex plane as the parameter k changes in the interval $(0, \infty)$. Is the critical point always stable?

Exercise 2.3.7. Given the ODE-system $\dot{\mathbf{u}} = A\mathbf{u}$, $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

Show, by solving the system that the solution at the critical points are unstable!

Hence when two eigenvalues $= 0$ (0 is a double root) the solution *can* be unstable.

Exercise 2.3.8. Discuss the stability concept for the inhomogeneous LCC-system

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = A\mathbf{u} + \mathbf{g}(t)$$

In what sense is a solution stable even if it is unbounded, i.e. $\mathbf{u}(t) \to \infty$ as $t \to \infty$? Study the simple ODE (sketch trajectories)

$$\frac{\mathrm{d}u}{\mathrm{d}t} = -u + t$$

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2.4 Some ODE-models in science and engineering

Mathematical models based on ODE-systems occur frequently in science and engineering. The independent variable usually denotes time t or space x.

A time dependent ODE-system is often referred to as a *dynamic system*. Since such a model does not take into account any space variations of the dependent variables it is sometimes called a *lumped* model. A space dependent ODE-system involves only one space variable but does not take time into account. Such models often occur by simplifying a stationary PDE-model in several space variables into a 1D space dependent problem.

In mathematical models of processes in science and engineering it is important to define the *units* of the variables and parameters, e.g. density ρ $[kg/m^3]$, flow Q $[m^3/s]$, gravitational acceleration g $[m/s^2]$. In this textbook SI-units will be used in most presentations of "nature laws", examples and exercises.

2.4.1 Newton's second law

A system of moving particles with masses m_1, m_2, ..., $m_n[kg]$ move according to:

$$m_i\ddot{\mathbf{r}}_i = \mathbf{F}_i(t, \mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_n, \dot{\mathbf{r}}_1, \dot{\mathbf{r}}_2, \ldots, \dot{\mathbf{r}}_n), \quad i = 1, 2, \ldots, n \tag{2.43}$$

Here $\mathbf{r}_i = (x_i, y_i, z_i)^T$ $[m]$ denotes the position of particle i, $\dot{\mathbf{r}}_i$ $[m/s]$ its velocity and $\ddot{\mathbf{r}}_i$ its acceleration $[m/s^2]$. $\mathbf{F}_i$ $[N]$ is the total force acting on particle i. The system (2.43) consists of $3n$ second order ODEs.

This equation was formulated by the English mathematician and scientist Isaac Newton in his work Philosophiae Naturalis Principia Mathematica (Mathematical Principles of Natural Philosophy) from 1687.

Engineering applications of (2.43) occur in e.g. mechanical vibrations. One important model consists of a system of masses connected by dampers and springs:

$$M\ddot{\mathbf{x}} + D_\nu\dot{\mathbf{x}} + K\mathbf{x} = \mathbf{F}(t) \tag{2.44}$$

Here M denotes the *mass* matrix, D_ν the *damping* matrix and K the *spring* or *stiffness* matrix. $\mathbf{x}$ is a vector of displacements of the masses along the x-direction. $\mathbf{F}$ is a vector of external forces often of the form $\mathbf{F}(t) = \hat{\mathbf{F}} \cdot \sin(\omega t)$, where ω is the angular velocity, see exercise 2.1.5 for an example. Observe that the ODE-system (2.44) is an LCC-system.

2.4.2 Hamilton's equations

A conservative mechanical system is defined as a system where the total energy is conserved. In such a system the motions can, as an alternative to Newton's

equations, be described by a scalar *hamiltonian* function H invented by the Irish mathematician W.R.Hamilton in 1835.

H is the sum of the kinetic enery T $[J]$ and the potential energy U $[J]$ of the system. For a system of n particles in 3D the hamiltonian has the form:

$$H = T + U = H(\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_n, \mathbf{p}_1, \mathbf{p}_2, \ldots \mathbf{p}_n), \tag{2.45}$$

where the $\mathbf{q}_i$ denote the generalized coordinates and $\mathbf{p}_i$ are the generalized momenta. Hamilton's equations of motion take the form

$$\frac{\mathrm{d}\mathbf{q}_i}{\mathrm{d}t} = \frac{\partial H}{\partial \mathbf{p}_i}, \quad \frac{\mathrm{d}\mathbf{p}_i}{\mathrm{d}t} = -\frac{\partial H}{\partial \mathbf{q}_i}, \quad i = 1, 2, \ldots, n \tag{2.46}$$

The ODE-system (2.45) consists of $6n$ first order ODEs.

2.4.3 Electrical networks

An electrical network consists of passive elements such as resistors, capacitors and inductors and active elements such as voltage and current sources. The model of a network is obtained by using Kirchhoff's laws and current-voltage relations due to Ohm's law for all elements. If all elements are assumed to have constant impedance values, a linear system of *differential-algebraic equations* (DAE-system) of the following form can be set up, with an algorithm called *modified nodal analysis*:

$$\begin{pmatrix} A_C C A_C^T & 0 & 0 \\ 0 & L & 0 \\ 0 & 0 & 0 \end{pmatrix} \frac{\mathrm{d}}{\mathrm{d}t} \begin{pmatrix} \mathbf{e} \\ \mathbf{i}_L \\ \mathbf{i}_V \end{pmatrix} + \begin{pmatrix} A_R G A_R^T & A_L & A_V \\ -A_L^T & 0 & 0 \\ A_V^T & 0 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{e} \\ \mathbf{i}_L \\ \mathbf{i}_V \end{pmatrix} = \begin{pmatrix} -A_I \mathbf{I} \\ \mathbf{0} \\ \mathbf{E} \end{pmatrix} \tag{2.47}$$

where C $[F]$, L $[H]$ and G $[\Omega^{-1}]$ are diagonal matrices with the capacitances, inductances and conductances (inverses of resistances), respectively A_C, A_R, A_L, A_V and A_I are (reduced) incidence matrices, i.e. sparse matrices (see appendix A.5) whose elements are the integers $-1, 0, 1$. The vector $\mathbf{I}$ $[A]$ consists of the current source values and $\mathbf{E}$ $[V]$ of the voltage source values.

The DAE-system (2.47) consists of differential equations for $\mathbf{e}$ $[V]$ (the node potentials) and $\mathbf{i}_L$ $[A]$ (vector of branch currents through the inductors) and additional algebraic equations for $\mathbf{i}_V$ $[A]$ (vector of branch currents through voltage sources).

Linear DAE-systems in general have the following form (special case of the DAE-system formulated in exercise 2.1.7):

$$A\dot{\mathbf{u}} = B\mathbf{u} + \mathbf{b}(t) \tag{2.48}$$

where A and B are $n \times n$-matrices. If A is *singular* the system has *index* ≥ 1: at least one differentiation is required to produce an ODE-system.

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2.4.4 Chemical kinetics

Consider a mixture of N chemical substances $A_1, A_2, \ldots, A_N$ reacting with each other in M reactions, where the substance concentrations change with time

$$\sum_{j=1}^{N} \alpha_{ij} A_j \rightarrow \sum_{j=1}^{N} \beta_{ij} A_j, \quad i = 1, 2, \ldots, M$$

where α_{ij} and β_{ij} are integer numbers called the *stoichiometric coefficients*. The time dependent substance concentrations $c_1, c_2, \ldots, c_N$ $[mol/m^3]$ of $A_1, A_2, \ldots, A_N$ are obtained from the ODE-system

$$\frac{\mathrm{d}\mathbf{c}}{\mathrm{d}t} = S\mathbf{r}(\mathbf{c}) \tag{2.49}$$

In (2.49) S is the stoichiometric matrix of integers, a sparse $N \times M$ matrix, and $\mathbf{r}(\mathbf{c})$ $[mol/(m^3 \cdot s)]$ an $M \times 1$ vector of reaction rates. In a model by the Norwegian chemists Guldberg and Waage from 1879 called the *mass action law*, r_i has the form

$$r_i = k_i \prod_{j=1}^{N} c_j^{\alpha_{ij}}, \quad i = 1, 2, \ldots, M \tag{2.50}$$

where k_i (the unit depends on N and α_{ij}) is called the rate constant which is temperature dependent according to Arrhenius' equation $k = Ae^{-E/RT}$, where A is called the pre-exponential factor, E $[J/mol]$ is the activation energy, $R = 8.314472$ $[J/(K \cdot mol)]$ the gas constant and T $[K]$ the temperature. Svante Arrhenius was a Swedish chemist and Nobel laureat active about 100 years ago.

The ODE-system (2.49) is generally nonlinear. With slight modifications it can be used to model chemical reactors of great importance in applications.

2.4.5 Control theory

A common model used in control theory is the *state-space* model

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x} + B\mathbf{u} \tag{2.51a}$$

$$\mathbf{y} = C\mathbf{x} + D\mathbf{u} \tag{2.51b}$$

where $\mathbf{x}$ are internal state variables of the system studied, $\mathbf{u}$ are control variables and $\mathbf{y}$ are observed variables. A, B, C and D are constant matrices. Of special interest is the study of *stability properties* of control systems. Another important problem in control theory is the *identification* problem, i.e. from measurements of the observed variables determine some of the parameters of the system.

In control theory the models are often descibed by *block diagrams* which can be translated into ODE-systems of type .

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Models in control theory can also be nonlinear ODE-systems of type

$$\frac{\mathrm{dx}}{dt} = \mathrm{f}(\mathrm{x}, \mathrm{u}, \mathrm{p}) \tag{2.52a}$$

$$\mathrm{y} = \mathrm{g}(\mathrm{x}, \mathrm{u}, \mathrm{p}) \tag{2.52b}$$

where $\mathbf{p}$ is a parameter vector.

2.4.6 Compartment models

Compartment models are frequently used in applications where simplified models of flows of gases or fluids are studied. Examples are found in e.g. biomedicine, pharmacokinetics and air pollution. The model consists of a system of first order ODEs, describing the exchange of substances between the compartments of the model. Assume the model has n compartments, which can be regarded as "boxes". If x_i $[mol]$ is the amount of some substance in compartment i, the model is formulated as an IVP, based on the balance principle for the amounts of the substances called the *continuity equation* (see chapter 9):

$$\frac{dx_i}{dt} = Q_i^I + \sum_{j=1}^{n} Q_{ij}^I - \sum_{j=1}^{n} Q_{ij}^O - Q_i^O, \qquad x_i(0) = x_{i0} \tag{2.53}$$

where the inflow consists of Q_i^I $[mol/s]$ which is the flow from the environment to compartment i and Q_{ij}^I the flow to compartment i from compartment j. The outflow consists of Q_i^O, which is the flow from compartment i to the environment and Q_{ij}^O being the flow from compartment i to compartment j. The initial value x_{i0} is the initial amount of the substance in compartment i.

If the flows are modelled as $Q_{ij}^I = k_{ij}^I x_j$, $Q_{ij}^O = k_{ij}^O x_i$ and Q_i^I, Q_i^O are constant or time dependent, the resulting ODE-system will be an LCC-system.

2.5 Some examples from applications

In this section of the chapter some ODE-applications are shown, exemplifying the models presented in 2.4. The examples will be referred to in the following chapters 3 and 4, where numerical methods for IVPs and BVPs are described. To each example some exercises are given. They are mainly intended for repetition of analytic treatment of ODEs but also to demonstrate analytic preprocessing necessary before numerical treatment of the problems should be performed.

Example 2.3 (Particle dynamics). The following differential equation is a model of the motion in the xy-plane of a particle released with a certain initial velocity from a point and then moving under the influence of gravity and a force generated

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by the air resistance which is proportional to the square of the velocity (see exercise 2.1.4 for the corresponding 1D problem).

$$m\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} = -mg\mathbf{e}_y - c\left\|\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}\right\|_2 \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}, \quad \mathbf{r}(0) = y_0\mathbf{e}_y, \quad \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}(0) = \mathbf{v}_0 \tag{2.54}$$

where m $[kg]$ is the mass of the particle, $\mathbf{r}$ $[m]$ is the position of the particle, g $[m/s^2]$ gravitational acceleration, c $[kg/m]$ air resistance coefficient, y_0 $[m]$ initial height of the particle and $\mathbf{v}_0$ $[m/s]$ the initial velocity vector, which has an elevation angle $\alpha > 0$ with the horizontal plane (see figure 2.5).

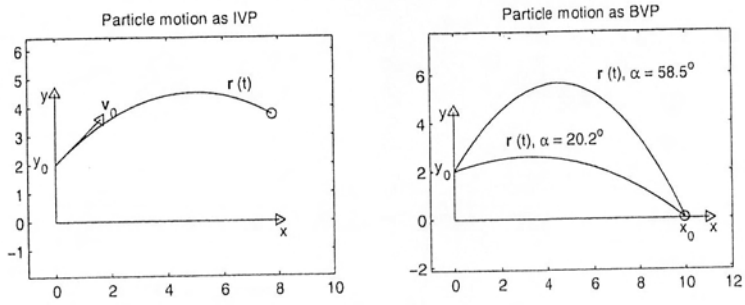


Figure 2.5: Particle dynamics in 2D

The formulation (2.54) of the problem is an IVP. The computational task is to integrate step by step forward in time until the particle hits the ground ($y = 0$). However, this problem can also be formulated as a BVP if the task is to determine the elevation angle that causes the particle to hit a given point x_0 on the ground for a given value of the initial velocity. The initial conditions are replaced by boundary conditions:

$$\mathbf{r}(0) = y_0\mathbf{e}_y, \quad \mathbf{r}(T) = x_0\mathbf{e}_x \tag{2.55}$$

where T is the time when the particle hits the ground in the point $(x_0, 0)$. By adjusting the elevation angle α the boundary condition at $x = x_0$ can be satisfied. It turns out that this BVP has two solutions, see the figure and chapter 4.

Exercise 2.5.1. Solve analytically the IVP and the BVP in the case $c = 0$.

Exercise 2.5.2. Rewrite the IVP as an ODE-system of first order, i.e. in the form (2.8).

Example 2.4 (Planetary motion). Consider the motion of a single planet about a heavy sun. The assumption heavy sun means that the sun influences the motion of the planet, but the relatively small mass of the planet has an influence on the sun that is negligible. Let $\mathbf{r}(t) = \big(x(t), y(t)\big)^T$ $[m]$ denote the coordinates of the planet with the sun at the origin. The only force acting on the planet is the gravitational force from the sun having the magnitude $F_r = \gamma Mm/\left\|\mathbf{r}\right\|_2^2$ $[N]$

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where $\gamma = 6.6742 \cdot 10^{-11}$ $[N \cdot m^2/kg^2]$ is the gravitational constant, M $[kg]$ the mass of the sun and m the mass of the planet.

Newton's equation (2.43) takes the form (for illustration, see figure 2.6):

$$m\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} = -\frac{\gamma M m}{\|\mathbf{r}\|_2^2}\mathrm{e}_r \tag{2.56}$$

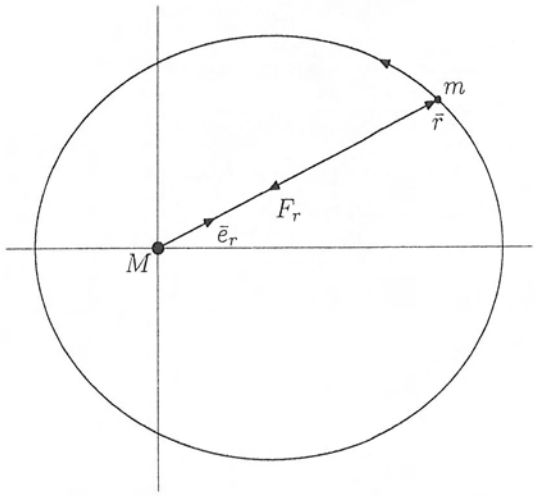


Figure 2.6: Planetary motion in 2D

Exercise 2.5.3. Write Newton's equation (2.52) in component form, i.e. one second order ODE for $x(t)$ another for $y(t)$. Also write (2.56) as a system of first order ODEs.

Exercise 2.5.4. For planetary motion the hamiltonian can be written

$$H(\mathbf{q},\mathbf{p}) = T + U = \frac{1}{2m}(p_1^2 + p_2^2) + \frac{\gamma M m}{\sqrt{q_1^2 + q_2^2}} \tag{2.57}$$

where $\mathbf{q}$ and $\mathbf{p}$ are the (generalized) coordinates and momenta

$$\mathbf{q} = \mathbf{r}, \quad \mathbf{p} = m\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$$

(a) Show that H is a constant, i.e. the total energy (sum of kinetic and potential energy) is conserved. Hint: Form $\mathrm{d}H/\mathrm{d}t$.

(b) Show that Hamilton's equations applied to (2.57) imply Newton's equation (2.56).

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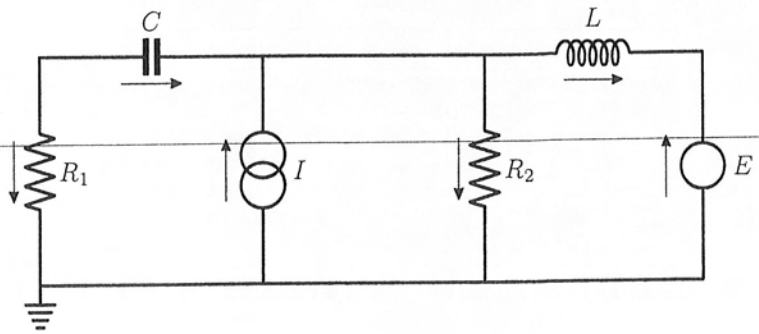


Figure 2.7: A simple electrical network

Example 2.5 (A simple RLC-circuit)**.** Consider the electrical network in figure 2.7, consisting of two resistors, one capacitance, one inductance, one voltage source and one current source.

The DAE-system corresponding to this electrical network is:

$$\begin{pmatrix} C & -C & 0 & 0 & 0 \\ -C & C & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & L & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \frac{\mathrm{d}}{\mathrm{d}t} \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ i_L \\ i_V \end{pmatrix} + \begin{pmatrix} \frac{1}{R_1} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{R_2} & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ i_L \\ i_V \end{pmatrix} = \begin{pmatrix} 0 \\ -I \\ 0 \\ 0 \\ E \end{pmatrix} \tag{2.58}$$

In a linear DAE-system there are linear relations between the dependent variables and the initial values must be consistent with these relations.

Exercise 2.5.5. Verify that the following linear algebraic relations are valid for a solution of the DAE-system above:

$$\begin{cases} R_1^{-1}e_1 + R_2^{-1}e_2 + i_L = -I \\ i_L = i_V \\ e_3 = E \end{cases}$$

and that the following initial condition is consistent with these algebraic relation: $\mathbf{u}_0 = (0, 0, E, -I, -I)^T$

Exercise 2.5.6. Find the steady state solution of (2.58).

Example 2.6 (Biochemical reaction)**.** The following reaction system, consisting of two reversible reactions, occurs frequently in biochemistry (the ESCEP-problem)

$$E + S \longleftrightarrow C \longleftrightarrow E + P$$

In the reaction formula, E stands for enzyme, S for substrate, C for complex and P for product. The respective concentrations are denoted by c_1, c_2, c_3 and

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c_4 $[mol/m^3]$. The ODE-system modelling the kinetic behaviour of the reaction system is

$$\frac{\mathrm{dc}}{\mathrm{d}t} = \begin{pmatrix} -1 & 1 & 1 & -1 \\ -1 & 1 & 0 & 0 \\ 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} k_1 c_1 c_2 \\ k_2 c_3 \\ k_3 c_3 \\ k_4 c_1 c_4 \end{pmatrix}, \quad \mathbf{c}_0 = \begin{pmatrix} E_0 \\ S_0 \\ 0 \\ 0 \end{pmatrix} \tag{2.59}$$

where the positive parameters k_1, k_4 $[m^3/s]$ and k_2, k_3 $[1/s]$ are the rate constants of the reactions and $E_0 > 0$ and $S_0 > 0$ are the initial concentrations of E and S.

Exercise 2.5.7. Verify that $c_1(t) + c_3(t) = E_0$ and $c_2(t) + c_3(t) + c_4(t) = S_0$.

When the reactions have come to chemical equilibrium, the concentrations satisfy the following nonlinear algebraic system of equations

$$\mathbf{f}(\mathbf{c}) = \begin{pmatrix} -1 & 1 & 1 & -1 \\ -1 & 1 & 0 & 0 \\ 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} k_1 c_1 c_2 \\ k_2 c_3 \\ k_3 c_3 \\ k_4 c_1 c_4 \end{pmatrix} = 0 \tag{2.60}$$

One solution of this system is obviously $\mathbf{c} = (0,0,0,0)^T$ which, however, cannot be the equilibrium concentrations since the relations in exercise 2.5.7 are not fulfilled. Using Newton's method (see appendix A.1) to solve the system of equations (2.60) will not work, since the matrix in (2.60) is singular which implies that the jacobian of $\mathbf{f}(\mathbf{c})$ also will be singular.

Exercise 2.5.8. Preprocess the equilibrium problem so that Newton's method can be used.

Example 2.7 (Stability of a regulator)**.** Consider a servo mechanism where the output signal $y(t)$ depends on an input signal $u(t)$. In an application $u(t)$ can be the torsion angle of the steering wheel of a boat and $y(t)$ the torsion angle of the boat's rudder. The purpose with a servo is to strengthen the torsion moment from a small value to a large value.

The following control system models a feedback regulator consisting of an electromechanical system described with a block diagram:

In mathematical terms the model is formulated in the following way:

$\oplus$: $e(t) = u(t) - y(t)$, the difference (error) between $u(t)$ and $y(t)$

G_1 : $v(t) = k_1 e(t)$, amplification of $e(t)$ into a voltage value

G_2 : $Ri(t) + L\frac{\mathrm{d}i}{\mathrm{d}t} = v(t)$, induced current in a motor driving an axis.

G_3 : $M(t) = k_2 i(t)$, torsion moment of the axis

G_4 : $J\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + d_\nu \frac{\mathrm{d}y}{\mathrm{d}t} = M(t)$, the moment drives a load (the rudder) having moment of inertia J and damping coefficient d_ν.

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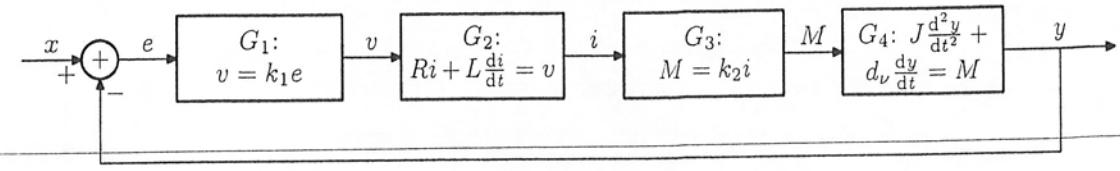


Figure 2.8: Block diagram of a servo mechanism

Exercise 2.5.9. Write the equations above in state space form (2.4.5) with $\mathbf{x}(t) = \big(i(t), z(t), y(t)\big)^T$, where $z(t) = \mathrm{d}y/\mathrm{d}t$, $\mathbf{u}(t) = \big(u(t), 0, 0\big)^T$ and $\mathbf{y}(t) = y(t)$. Hint: Eliminate $e(t), v(t)$ and $M(t)$. Note that the resulting ODE-system is of LCC-type.

Exercise 2.5.10. Verify that the equations above can also be written as a third order ODE (of LCC-type):

$$\frac{LJ}{R}\frac{\mathrm{d}^3y}{\mathrm{d}t^3} + \left(\frac{Ld_\nu}{R} + J\right)\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + d_\nu\frac{\mathrm{d}y}{\mathrm{d}t} + Ky = Ku(t), \quad K = \frac{k_1k_2}{R} \tag{2.61}$$

The properties of this regulator model can be simulated in a parameter study. All the parameters are positive. Particularly important is the parameter K, which is proportional to the amplification factors of the model. Will the model be stable for positive values of K? To answer this question it is sufficient to study the homogeneous version of (2.61), i.e. the case where $u = 0$.

$$\frac{LJ}{R}\frac{\mathrm{d}^3y}{\mathrm{d}t^3} + \left(\frac{Ld_\nu}{R} + J\right)\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + d_\nu\frac{\mathrm{d}y}{\mathrm{d}t} + Ky = 0 \tag{2.62}$$

If the parameter $L = 0$, (2.62) is a second order ODE of the same type as in exercise 2.3.6 and therefore stable for all values of K. If $L \neq 0$, however, we need to study the eigenvalues of (2.62) as function of K, and see if they stay in the left complex plane, or if they move over in the right complex plane. see exercise .

Exercise 2.5.11. Assume the following values of the parameters: $J = 0.4$, $d_\nu = 1$, $R = 100$, $L = 10$. Write a program that makes a graph (root locus) of the curves generated by the eigenvalues of (2.62) as K varies in the interval $[0, 20]$. For which values of K is the system unstable?

Example 2.8 (Drug kinetics)**.** The following compartment model is an example from pharmacokinetics. Assume a drug, which has been taken orally, is present in the intestine during a certain time interval. The drug is absorbed with the constant flow rate q $[mmol/s]$ into the first compartment, the blood plasm. In the blood plasm the concentration of the drug is $c_1(t)$ $[mmol/l]$. The second compartment is the organ where the drug is active. Between the first and second compartment there is an drug exchange with rate $k_1c_1(t)$ $[mmol/s]$ leading to the drug concentration $c_2(t)$ in the second compartment. In the organ the drug is

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consumed with the rate $k_b c_2(t)$ $[mmol/s]$ and the surplus is sent back to the blood with the rate $k_2 c_2(t)$ $[mmol/s]$. From the blood, finally, there is an elimination of the drug through the kidneys with the rate $k_e c_1(t)$ $[mmol/s]$. The following figure illustrates the blood-organ compartment model of drug distribution:

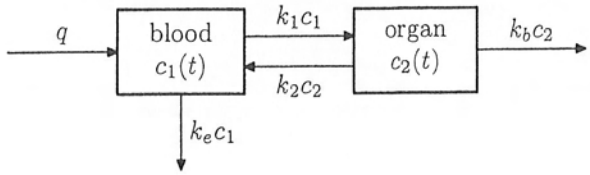


Figure 2.9: A simple compartment model

The ODE-system (2.53) takes the following form in this case:

$$\begin{aligned} \frac{dc_1}{dt} &= -k_1 c_1 + k_2 c_2 - k_e c_1 + q, & c_1(0) &= 0 \\ \frac{dc_2}{dt} &= k_1 c_1 - k_2 c_2 - k_b c_2, & c_2(0) &= 0 \end{aligned}$$

The steady-state solution, i.e. the solution obtained when the time derivatives are zero, is obtained from the algebraic linear system of equations:

$$\begin{pmatrix} k_1 + k_e & -k_2 \\ -k_1 & k_2 + k_b \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} q \\ 0 \end{pmatrix}$$

Exercise 2.5.12. Formulate the analytical solution of the ODE-system using Duhamel's formula (2.25).

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